

# Colin Vu

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## EDUCATION

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**Georgia Institute of Technology** Atlanta, GA

**M.S. in Computer Science**

August 2026 – May 2027

- Concentration: Machine Learning
- Coursework: Natural Language Processing, Computer Vision, Robot Reinforcement Learning, High Performance Computing

**B.S. in Computer Science, Minor in Industrial Design**

August 2022 – May 2026

- GPA: 3.94; Concentrations: Media/Graphics & Modeling/Simulation
- Coursework: Algorithms, Data Structures, Computer Architecture, Computer Networks, Computer Simulation, Human Centered Design, Computer Graphics & Animation, Machine Learning, GIS, Data Visualization

## SKILLS AND AWARDS

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- Languages: Python, JavaScript, TypeScript, Java, C++, C#, C, HTML, CSS, SQL, Swift
- Libraries & Frameworks: React, Node, LangChain, PyTorch, OpenGL, RxJS, React Query, Redux, OpenCV, Pixi, Scikit-Learn, TailwindCSS, Whisper, HTK, MediaPipe, CLIP, DINOv2
- Tools: Git, Cursor, Docker, AWS (EC2, S3, Route 53), Jira/Confluence, Figma, ArcGIS, Blender, Mixpanel, Datadog
- Awards/Achievements: HackGT12 Winner, Bronze Palm Eagle Scout, Tau Beta Sigma Affiliate

## EXPERIENCE

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**Talos Trading** New York, NY

May 2026 – August 2026

**Frontend Software Engineering Intern**

- Shipped 46 Jira tasks (80 PRs, 22K+ lines) across portfolio management, trading, and configuration workflows using React, TypeScript, RxJS, and Redux; many were requested directly by enterprise clients, spanning both new features and bug fixes.
- Designed and implemented product features alongside product and design teams, presented in 2 company-wide demos; wrote hundreds of Cypress and Vitest tests and integrated REST and WebSocket APIs with backend engineers.
- Resolved client-reported production issues, independently investigating root causes with Datadog; identified and fixed 15+ bugs, including regressions, flaky tests, migrations, and validation errors.

**Seraphine Glass** San Francisco, CA

April 2025 – August 2025

**Founding Engineer**

- Developed augmented-reality glasses with a built-in AI remembrance agent/lifestyle manager; integrated LangChain, Whisper, Google Places, NWS, and GPT with device inputs and external data sources. Deployed backend on AWS (EC2, S3, Route 53).
- Secured \$155,000 in pre-seed funding from Fusen and the Georgia Tech Startup Exchange.
- Pitched to 10+ VCs to discuss funding; negotiated hardware needs with 5+ manufacturers to develop a supply chain. Built TikTok and Instagram pages for marketing content, averaging 1,000+ organic views per video.

**American Gaming Systems** Duluth, GA

May 2024 – August 2024

**Fullstack Software Engineering Intern**

- Developed a client-facing UI for casino managers to modify slot machine game assets, programmed in Pixi, Node.js, and React.js. Wrote unit tests integrated with C++ RESTful APIs. Delivered project demos to business team and CTO.

## PROJECTS & RESEARCH

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**Symbiotic AI (Contextual Computing Lab, ICASSP 2027 Submission)**

September 2025 – Present

- First-author computer vision paper showing that handheld warehouse objects can be classified in real time without human labeling.
- Detected hand-state changes in head-worn display video using ArUco markers, MediaPipe landmarks, and DINOv2 embeddings; a CNN reached 96% state-change accuracy, with a weakly supervised HMM extension in progress.
- Designed a weakly supervised simulated annealing algorithm that infers handheld objects in videos from head-worn displays based on frame-by-frame CLIP embeddings. Achieved 99.7% object assignment accuracy and 97.9% inference accuracy.

**Field Goal Projection (HackGT12 Winner)**

September 2025

- Built a CV system that tracks a football in TV footage and projects its 3D coordinates (calibrating for camera movement) to calculate field goal distance. Integrated CSRT tracking, Kalman filtering, HSV isolation, and homography-based calibration.

**Factxis (Friendly Cities Research Lab)**

September 2025 – Present

- Developed a geographic data visualization tool (JavaScript & D3) for researchers to analyze socioeconomic indicators based on any dataset, with exploration across percentile-based views. Presented to and adopted by the Atlanta Regional Commission (ARC).

**GeoAI (Georgia Tech Research Institute)**

August 2023 – December 2024

- Built geographically weighted regression (GWR) models on federal datasets, standardized in MySQL, to analyze spatial variation.